Expanding the squared sum we see that there are (i) n squares J_k^2 plus (ii) products of the form $\mathrm{J}_k\mathrm{J}_l$, $k \neq l$. Clearly the former are always one and the latter have zero expectation, thus

$$\mathbb{E}\mathrm{X}_n^2 = n\delta x^2.$$

The variance therefore grows linearly with the number of steps taken.

Now let us take a slightly different perspective. We start with a time interval $[0, \tau]$ and subdivide this into ν steps of size $\delta t = \tau/\nu$. Let $t_n = n\delta t$, $n = 0, 1, \ldots, \nu$ and define

$$\mathrm{Y}(t_{k+1}) = \mathrm{Y}(t_k) + \sqrt{\delta t}\mathrm{J}_k, \qquad k = 0, 1, \ldots, \nu - 1,$$

where $\mathrm{Y}(0) = 0$. Effectively we are making the size of the jump δx taken at each step of the random walk equal to $\sqrt{\delta t}$.

We will use the shorthand $\mathrm{Y}_k = \mathrm{Y}(t_k)$ and $\Delta\mathrm{Y}_k = \mathrm{Y}_{k+1} - \mathrm{Y}_k = \sqrt{\delta t}\mathrm{J}_k$.

The increments are clearly independent. Let us observe the following facts, for $0 \leq k \leq l \leq \nu$

(i) $\mathbb{E}(\mathrm{Y}_l - \mathrm{Y}_k) = 0$,
(ii) $\mathrm{Var}(\mathrm{Y}_l - \mathrm{Y}_k) = t_l - t_k$.

For points between t_k and t_{k+1} we define the value of $\mathrm{Y}(t)$ by linear interpolation:

$$\mathrm{Y}(t) = \mathrm{Y}_k + \frac{(t - t_k)}{\delta t}\Delta\mathrm{Y}_k$$

Then, for $0 \leq t \leq s \leq \tau$, we have

$$\mathbb{E}(\mathrm{Y}(s) - \mathrm{Y}(t)) = 0.$$

This is easy to demonstrate. However, the intuitive corresponding statement $\mathrm{Var}(\mathrm{Y}(s) - \mathrm{Y}(t)) = (s - t)$ does not quite hold due to the reliance on interpolation. Nevertheless, such a relation holds in an approximate sense and in the limit of small step.

The relation $\delta t = \delta x^2$ defines a scaling of time in relation to distance. When we take limits as these two quantities go to zero in such a way that $\delta t = \delta x^2$ we refer to it as a *diffusion limit*. It is this type of random diffusion model that has the greatest relevance for experiments like those of Brown and Perrin.

In the diffusion limit the motion of the random walk can be viewed as an everywhere continuous (but almost nowhere smooth) path which is called a *Wiener process* (after Norbert Wiener).

$$\mathrm{W}(t) = \widehat{\lim}_{\delta t \to 0} \sqrt{\delta t} \sum_{k=0}^{\lfloor t/\delta t \rfloor} \mathrm{J}_k.$$